SUPERSEDING
MIL-A-23773 (Wep)
25 June 1963

MILITARY SPECIFICATION

ANCHOR, UNDERWATER MINE, MARK 56 MOD 4 AND MOD 5

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Anchor, Underwater Mine, Mark 56 Mod 4 and Mod 5, 549694 and 549718. They are of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications or such portions thereof as designated herein of the issue in effect on the date of invitation for bids form a part of this document.

SPECIFICATIONS

MILITARY

MIL-C-21947	Compound, Antifouling, For Underwater Cables, Navy Formula No. 184
MIL-G-7711	Grease, Aircraft, General Purpose

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 479467	Shipping Pallet for Mine Anchor Mark 56 Type
LD 549694	Anchor, Underwater Mine, Mark 56 Mod 4
LD 549718	Anchor, Underwater Mine Mark 56 Mod 5

CLASSIFICATION OF DEFECTS

CD 2421432	Anchor, Underwater Mine, Mark 56 Mod 4
CD 2421477	Anchor, Underwater Mine, Mark 56 Mod 5

And all specifications, standards and drawings listed thereon.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The Anchor, Underwater Mine, Mark 56 Mod 4 or Mod 5, herein after referred to as the "mine anchor", is a weight, including a cable pay out device disengageable from a mine after launching and operable to anchor the mine at a predetermined depth below the surface of the water.

3.2 General requirements.

3.2.1 Preproduction sample. - A preproduction sample of the mine anchor shall be manufactured using the methods and procedures proposed for the production lot. The preproduction sample shall be prepared for delivery in accordance with Section 5. This sample will be inspected as specified in Section 4 herein and is for the purpose of determining that the production meets the requirements of this specification and the design. Provision of the sample shall be in accordance with 4.3.1.

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3.2.2 Fabrication. - The mine anchor shall be manufactured in accordance with NOSC LD 549694 or 549718, all documents listed therein, and as specified herein. Inspection and acceptance shall be in accordance with Section 4.

3.2.3 Conflicting requirements. - Conflicting requirements arising between this specification and any other specification, publication, or drawing listed herein shall be referred to the procuring agency or appointed agent in writing for interpretation, clarification and resolution.

3.2.4 Data. - No data is required by applicable documents referenced in Section 2, unless specified in the contract or order.

3.3 Performance requirements and product characteristics. - The mine anchor and its components and assemblies shall meet the following performance requirements and product characteristics:

3.3.1 Sub-assembly requirements.

3.3.1.1 Tube and flange assembly leakage. - The tube and flange assembly (Dwg. 2428478) shall exhibit no leakage when tested in accordance with 4.5.2.1.

3.3.1.2 Safety device assembly leakage. - The safety device assembly (Dwg. 1281099) shall exhibit no leakage when tested in accordance with 4.5.2.2.

3.3.2 Hold-off nut disengagement. - The hold-off nut shall be threaded on the drum shaft so that, when the drum shaft is positioned with its threaded end protruding the minimum amount through the anchor housing (Dwg. 1362676), eight (8) to nine (9) turns of the drum will cause the hold-off nut (Dwg. 1388051) to disengage from the finger (Dwg. 1388040).

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3.3.3 Finger assembly movement. - As the nut moves free of the drum shaft, by rotation of the cable drum, the finger assembly (Dwg. 1388038) shall move free of the pawl weldment (Dwg. 2419554).

3.3.4 Running in. - The drum, as wound with cable and assembled within the anchor, shall rotate upon its shaft so as to successfully complete the running-in test of paragraph 4.5.2.6.

3.3.5 Pull Out Test. - The force required to pull the cable out of the cowtail shall be no greater than 100 pounds to successfully complete the pull out test of paragraph 4.5.2.8.

3.3.6 Antifouling coating. - The entire length of cable shall be coated with antifouling compound in accordance with the provisions of 4.5.2.3.

3.3.7 Materials, finishes, coatings, construction, and dimensions. - Materials, finishes, coatings, construction, and dimensions for the mine anchor shall meet the requirements of LD 549694 or LD 549718 and this specification. Examination shall be in accordance with 4.5.1.

3.4 Durability requirements.

3.4.1 Pressure. - The container assembly (Dwg. 2428484) of the mine anchor shall be capable of withstanding the pressure test of 4.6.1.

3.5 Clean-up. - Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of metal chips and other foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of plating, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

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The standard of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall determine the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - Inspection of the items covered by this specification shall be classified as follows:

a. Preproduction inspection (See 4.5 and 4.6)

b. Quality conformance inspection (See 4.5)

c. Periodic production inspection (See 4.6)

4.3 Inspection.

4.3.1 Preproduction sample. - A preproduction sample of ten (10) mine anchors, manufactured in accordance with 3.2.1, shall be subjected to the examinations and tests detailed in 4.5 and 4.6 at an activity designated by the procuring activity. Acceptance of the sample shall be based on no defects in this sample. Further production of the mine anchor by the supplier, prior to approval of the sample, shall be at the supplier's risk.

4.3.2 Quality conformance.

4.3.2.1 Sampling. - Tests designated as sample tests in 4.5.2 shall be conducted on a random sample of

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mine anchors selected from each production lot (see 6.2.1) in accordance with the provisions of MIL-STD-105, Inspection Level II, AQL 0.65. Failure of a unit to pass any quality conformance sampling test shall result in rejection of that unit and shall result in rejection of the lot if the number of defectives exceeds the acceptance number for the stated AQL. No lot shall be shipped until the samples representing the lot have successfully passed the quality conformance sampling tests.

4.3.3 Periodic sampling tests. - After successful completion of the quality conformance inspection of 4.5, the periodic sampling tests of 4.6 shall be conducted on a random sample of five (5) mine anchors selected by the Government from each lot of mine anchors offered for acceptance. When a unit fails to pass any periodic sampling test or specified subsequent quality conformance test, examination of the unit shall be made by the contractor to analyze the cause of the failure and corrective action to be taken. When cause of the failure is determined to be supplier non-conformance to the quality conformance test requirements of this specification and/or its associated drawings, the contractor may be required to screen the lot and repair or replace the discrepant part or material and resubmit the lot for complete retesting.

4.4 Special test equipment. - The following items of special test equipment are required to perform the quality conformance inspection specified in 4.5:

a. Cable pay off test equipment, see Figure 1 (4.5.2.7)

b. Decimal second timing device. (4.5.2.7)

c. Adapter for attaching of tube weldment to air line. (4.5.2.1)

d. Housing leakage test closure. (4.5.2.2.1)

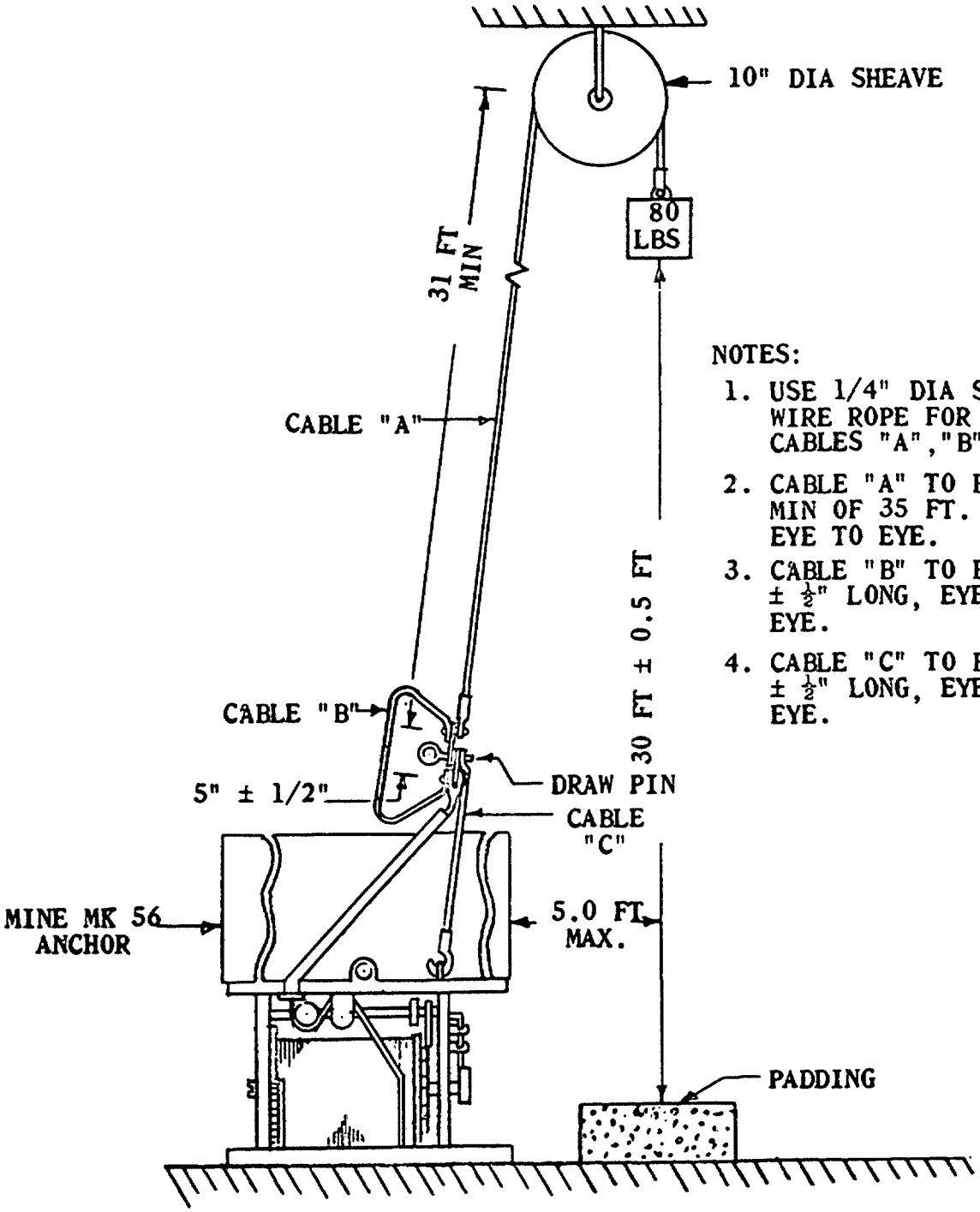
e. Piston leakage test closure. (4.5.2.2.2)

4.5 Quality conformance inspection.

4.5.1 Examination. - Each mine anchor shall be visually and dimensionally examined in accordance with CD 2421432 or 2421477 to determine compliance with this

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CABLE PAY OFF AND PAWLING TEST



NOTES:

1. USE 1/4" DIA STEEL WIRE ROPE FOR CABLES "A","B","C"
2. CABLE "A" TO BE A MIN OF 35 FT. LONG EYE TO EYE.
3. CABLE "B" TO BE 17 ± ½" LONG, EYE TO EYE.
4. CABLE "C" TO BE 25 ± ½" LONG, EYE TO EYE.

FIGURE 1

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specification and LDs 549694 and 549718. Examination in accordance with the CD shall not relieve the contractor of his responsibility to meet all drawing requirements not listed on the CD.

4.5.2 Tests.

4.5.2.1 Tube and flange assembly leakage test (Sampling). - Close and seal the amphenol plug ends of the tube and flange assembly (Dwg. 2428478) using plug (Dwg. 1827589) and a replaceable "0" ring (MS 29513-212). With a replaceable "0" ring (MS 29513-220) in the groove of the flange (Dwg. 2428478), attach the flange to an adapter of an air line whereby a controlled internal pressurization of the tube and flange assembly can be accomplished. The tube and flange assembly shall be pressurized to 10 ± 5 psig and submerged in water (including the adapter end) for 10 minutes. Any tube and flange assembly showing any evidence of leakage shall be rejected.

4.5.2.2 Safety device assembly leakage test.

4.5.2.2.1 Housing leakage test (Sampling). - The housing (Dwg. 1281098) shall be tested for leakage and porosity using a testing closure conforming to the piston (Dwg. 1489887) containing a threaded hole, in either the flange or stem, to which an air line can be attached. Place a replaceable "0" ring (AN 6227-33) on the flange of the testing closure, insert closure into housing, assemble a replaceable retaining ring (MS 16625-2256), and connect to air line. The entire testing assembly, housing and testing closure, shall be submerged in water and internally pressurized to 15 ± 5 psig for 10 minutes. Any housing showing evidence of leakage shall be rejected.

4.5.2.2.2 Piston leakage test (Sampling). - The piston (Dwg. 1489887) shall be tested for leakage using a testing closure conforming to the internal dimensions of the housing (Dwg. 1281098) containing a threaded hole in one end to which the air line can be attached. Place a replaceable "0" ring (AN 6227-33) on the flange of the piston, insert piston into testing closure, assemble a replaceable retaining ring (MS 16625-2256) to closure and connect to air line. The entire testing assembly, piston and testing closure, shall be submerged in water and

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internally pressurized to 15 ± 5 psig for 10 minutes. Any piston showing evidence of leakage around or through the piston shall be rejected.

4.5.2.3 Antifouling compound (Sampling). - Apply compound at 180° to 212° F. (do not overheat). Use a double boiler, with boiling water vented to the atmosphere in the bottom section. Do not use any type of oil or any water solution. Use only plain tap water. Do not remove the preservative compound from the cable before applying the antifouling compound. Move the cable slowly through the heated compound and then through a tapered wiper die .171" + .003" - .000" diameter. This should force the compound into the interstices of the cable and leave a thin film remaining on the outside surfaces of the cable. The cable shall be examined to assure that the entire surface of the cable has been thoroughly coated. Any section where the antifouling compound has not penetrated the interstices of the cable and a thin film does not remain on the surface of the cable shall be cause for rejecting the entire cable.

WARNING: Overheating of the antifouling compound can have deleterious effects on it and can also cause the release of toxic vapor.

4.5.2.4 Hold-off nut disengagement test (Sampling).-With the hold-off nut free of the drum shaft, engage the nut and manually thread the nut on the shaft nine (9) full turns. In a reverse manner remove the nut. During this operation any mine anchor in which the nut binds or sticks upon the threaded portion of the drum shaft, shall be regarded as defective.

4.5.2.5 Finger assembly movement test (Sampling). - With the hold-off nut removed from the drum shaft, rotate the drum. Determine that the finger of the finger assembly moves free of the pawl weldment. Any finger not moving free of the pawl weldment shall be cause for regarding the anchor as defective.

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4.5.2.6 Running-in test (Sampling). - With the brake tooth engaging the brake end of the drum, secure the free end of the cable to the drum in any suitable manner in which the drum may be rotated in either direction upon its shaft. With the cable so secured, rotate the drum a minimum of ten (10) complete revolutions in each direction. Any mine anchor in which any degree of binding or jamming of the drum occurs shall be regarded as defective. (See paragraph 4.5.2.9)

4.5.2.7 Cable pay off and pawling test (Sampling). - Position the hold-off nut so that two full turns of the cable drum will disegage the nut from the finger assembly. Arrange and secure the test cables to the anchor cables by a method that will, upon testing, effect a smooth release. Criteria for cable lengths and other critical dimensions are shown in the schematic drawing of Figure 1. The release mechanism shall accept the load of the 80 pound weight until release is effected. Upon release, time the descent of the weight through its 30 foot drop. Should the weight drop in less than 4.0 seconds or more than 8.0 seconds, or if the pawl fails to operate in locking the drum as slack develops in the cable, the mine anchor shall be regarded as defective and appropriate corrective measures shall be taken. (See paragraph 4.5.2.9)

NOTE: Possible points of malfunction may be: (1) a jammed brake due to misalignment or, (2) incorrectly installed torsion spring.

4.5.2.8 Pull out test (Sampling). - With the anchor assembled as shown in Figure 1, but without the three test cables, attach a suitable spring scale to the free end of the mooring cable. Pulling the cable slowly and vertically out of the cowtail until the drum has made at least one full revolution, the force measured shall be no greater than 100 pounds. Any anchor where the force exceeds 100 pounds shall be regarded as defective. (See paragraph 4.5.2.9)

4.5.2.9 Greasing of drum (Sampling). - After the tests of 4.5.2.6, 4.5.2.7 and 4.5.2.8, thoroughly coat the lobes of the brake end of the drum (Dwg. 2292982) with Grease, Aircraft, General Purpose - MIL-G-7714.

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4.6 Durability test: -

4.6.1 Pressure. - The container assembly Dwg. 2428484 suitably plugged and sealed shall be subjected to 545 ± 10 psig external hydrostatic pressure for a period of 5 minutes. Any container assembly showing evidence of leakage shall be regarded as defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. (Not applicable)

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. - Each mine anchor shall be individually packed in accordance with LD 479467.

5.2.2 Level B. (Not applicable)

5.2.3 Level C. (Not applicable)

5.3 Marking.

5.3.1 Normal markings. - In addition to any special marking required by the contract or order, all marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should include the following:

a. A Government activity or contractor, as appropriate, to perform preproduction and periodic sampling tests.

b. When preproduction samples are waived because of having been submitted on a recent contract, the standards of workmanship exhibited by the previously submitted preproduction sample shall determine the minimum requirements of the current contract relative to workmanship.

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c. If the contract establishes periodic production sample sizes at variance with this specification, the contract shall also furnish a new test plan, or in some manner designate the desired level and sequence of testing of each of the sample units.

d. Special test equipment to be loaned by the Government and the terms under which it is loaned.

e. Title, number and date of this specification.

f. Include the following only when the individual component is shipped directly to the Government or furnished as a replacement part. This need not apply when the component is purchased as part of a fully or partially assembled mine.

(1) Packaging level "A"

(2) Special markings, if required

g. When mine anchors are purchased from a prime contractor for installation into Assembly Condition "D" mine, preparation for delivery shall be adequate to protect the unit prior to installation.

h. If one month's scheduled production is not an acceptable lot size, then lot size should be specified by the contract. If this occurs, the sample size of 4.3.3 shall be modified accordingly.

6.2 Definitions. - The following terms will be used throughout this specification on the mine anchor as covered by LDs 549694 and 549718.

6.2.1 Lot size. - Unless otherwise specified in the contract or purchase order, lot size is defined as one month's scheduled production.

6.2.2 Mine anchor. - Reference to the "mine anchor" shall mean the Anchor, Underwater Mine, Mk 56 Mod 4 or Mod 5.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-A-23773A(OS) ANCHOR, UNDERWATER MINE, MARK 56 MOD 4 and MOD 5

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1426, APR 63

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Naval Ordnance Systems Command
Washington, D.C. 20360

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